Don Bosco Institute of Technology

An Autonomous Institute Affiliated to University of Mumbai,

Department of Computer Engineering

(Session 2026-27 Odd)

CSP701:Major Project-I
Review-I

Group No. - 01

Team Members:

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Proiect Guide:

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24th Aug, 2026

Literature Survey with summarized findings/ Gap Identification

1	**Mathematical model and method of decisionmaking in economic systems based on deep learning**	The paper introduces an interval-stochastic Analytic Hierarchy Process (AHP) A two-hidden-layer neural network was integrated to predict expected ROI from loan and borrower criteria, combining AI-driven prediction with the structured AHP ranking for P2P lending decisions
2	**AI-Based Fraud Prevention Models for SecuringReal-Time Financial Data Across LendingPlatforms**	Modeling users, transactions, devices, and IP addresses as nodes in a dynamic heterogeneous graph allowed the GNN to capture relational and temporal fraud patterns that traditional isolated-data-point methods miss.
10	**Artificial Intelligence in Credit Risk Management of Peer-to-Peer Lending Financial Technology:**	explainable AI (XAI) is more effective than black-box models for identifying, analyzing, and evaluating credit risk in P2P lendingLIME (Local Interpretable Model-Agnostic Explanations) and SHAP (Shapley) Value emerged as the most widely used machine learning frameworks for explaining credit score predictions, helping identify which borrower features (e.g., loan purpose, total interest received, recoveries) drive default risk classifications
9	**Imbalance-Aware Credit Default Risk Estimation for Peer-to-Peer Digital Lending Platforms via Dual-Level XGBoost Optimization with Focal Loss and Explainable AI**	the paper's proposed dual-level imbalance correction framework (XGBoost + SMOTE + Focal Loss + TreeSHAP) is the first to formalize this approach, improving sensitivity by 12–18% while maintaining exact TreeSHAP explainability for GDPR compliance.

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15	**Leveraging Artificial Intelligence to Predict Unfunded Loans in Peer-to-Peer Lending Platforms**	The study combined LR, RF, and SVM into a Heterogeneous Ensemble (HEE) model to predict investor funding outcomes on 105,666 SMOTE-balanced Prosper P2P loans (2018–2022), finding it significantly outperformed all individual classifiers (98% accuracy, 97% recall, 0.998 AUC) with statistically confirmed superiority ($p < 0.01$), making it the only model besides standalone RF to meet real-world suitability thresholds.
4	**Enhancing Financial Predictions with LSTM and LIME: A Cognitive Decision-Making Model Approach**	An LSTM+LIME model for P2P loan default prediction achieved strong, well-generalized results (89% training / 85% testing accuracy, F1-score 0.835), with LIME identifying loan amount and interest rate as the top default predictors combining accuracy with the transparency finance stakeholders need.
8	**Enhancing Credit Risk Prediction through an Ensemble of Explainable Model**	The paper shows that combining SHAP, LIME, and XGBoost improves the stability and interpretability of credit-risk predictions, particularly with imbalanced data. However, it lacks integration with blockchain, smart contracts, and decentralized lending, leaving a gap for future research.
7	**Machine Learning XAI for Early Loan Default Prediction**	The paper finds that XGBoost provides strong loan-default prediction, while surrogate decision trees and fuzzy linguistic rules improve interpretability through five understandable risk categories. However, it does not integrate the AI-based credit assessment with blockchain, smart contracts, or decentralized lending, leaving a gap for future research.

Introduction to Topic

Literature Survey with summarized findings/ Gap Identification

30	**Data-Driven Loan Default Prediction: A Machine Learning Approach for Enhancing Business Process Management**	The study combined LR, RF, and SVM into a Heterogeneous Ensemble (HEE) model to predict investor funding outcomes on 105,666 SMOTE-balanced Prosper P2P loans (2018–2022), finding it significantly outperformed all individual classifiers (98% accuracy, 97% recall, 0.998 AUC) with statistically confirmed superiority ($p < 0.01$), making it the only model besides standalone RF to meet real-world suitability thresholds.
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3	**Design Smart Contract based on Blockchain for P2P Lending Platform Decentralized Trustworthiness Score Management with Smart Contracts (TrustLend)**	Highly relevant to the proposed project as it is the closest match in domain and technology combined — it directly designs a blockchain and smart-contract architecture for a P2P lending platform, providing a conceptual system flow (registration, loan matching, smart contract generation, automated repayment) that can serve as a foundational reference for the system design/architecture chapter of the major project
19	**Smart e-KYC: a Privacy-Protected Consortium Blockchain-Based Framework for IoT-Enabled e-KYC System**	This paper provides a highly relevant architectural reference for secure, privacy-preserving user verification by combining on-chain hashing with encrypted off-chain IPFS storage to protect sensitive KYC documents. While it offers strong strategies to prevent identity fraud and mitigate vulnerabilities like borrower-recommender collusion, its scope is limited strictly to identity and data management; it does not address the core financial mechanics of P2P lending, such as credit scoring, loan default prediction, or automated lending decisions.
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Problem statement

- **Traditional banks create friction at every step, slow approvals, heavy paperwork, and rigid CIBIL-based assessments shut out freelancers and first-time borrowers from fair credit access.**

Key Challenges:

- Slow approvals & heavy paperwork
- Strict CIBIL scores exclude freelancers
- Banks retain large interest margins
- Low returns for depositors
- Limited transparency in lending
- No recourse for underserved borrowers
- Centralized risk with no audit trail

How BorroWealth Helps:

- The project aims to design a decentralized lending platform that removes intermediaries and uses AI-driven alternative credit scoring to enable secure, transparent, and inclusive financial transactions.

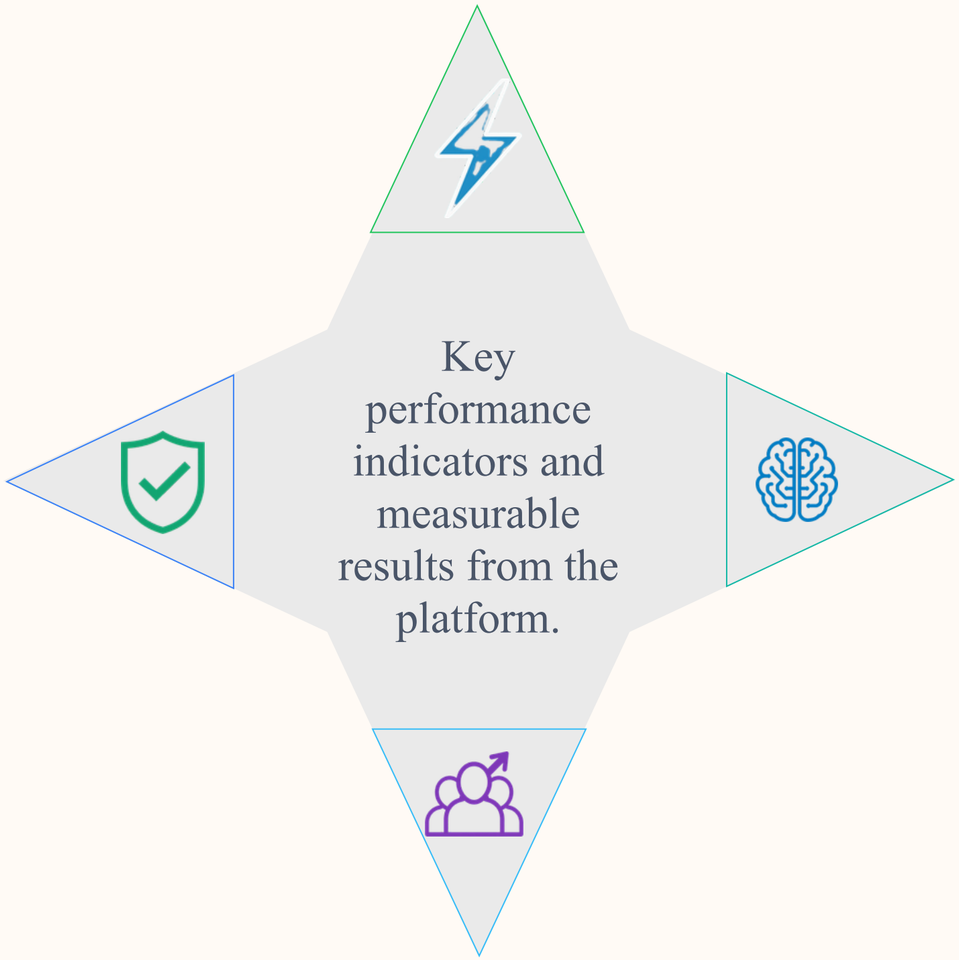
Objectives and Scope

Fast Approvals & Transparency

Loan approvals in under 5 seconds with fully transparent, auditable lending records on-chain.

Reduced Fraud & Automation

Smart contract automation eliminates manual processing and significantly reduces fraud risk.



AI Accuracy & Scale

Accurate AI credit scoring with simulated transactions and test users validated.

Financial Inclusion & Recovery

Credit score based reduction in defaults with alternative credit access.

Objectives & Scope



BORROWEALTH – SYSTEM ARCHITECTURE

AI-Powered Peer-to-Peer Lending Platform with Blockchain Transparency

USER LAYER

BORROWER

- Register / Login
- KYC Verification
- Loan Application
- AI Risk Score & Insights
- Loan Status
- Repayments

LENDER / INVESTOR

- Register / Login
- KYC Verification
- Browse Loans
- Filter & Search
- AI Risk Insights
- Invest / Fund Loan
- Portfolio Dashboard
- Repayments Received

ADMIN

- User Management
- KYC Monitoring
- Loan Monitoring
- Risk & Fraud Alerts
- System Analytics
- Dispute Management

1. PRESENTATION LAYER (WEB / MOBILE APPS)

- Borrower App
- Lender / Investor App
- Admin Dashboard

2. API GATEWAY

API Gateway / Load Balancer – Routes requests to appropriate microservices

3. BACKEND MICRO SERVICES LAYER

Authentication Service

- Register / Login
- JVC / Token Mgmt
- Password Reset
- Role Based Access

User Service

- User Profiles
- KYC Management
- Documents
- User Preferences

Loan Service

- Create Loan
- Update Loan
- Loan Lifecycle
- EMI Schedule
- Loan Status

Marketplace Service

- Publish Loans
- Search / Filter
- Loan Matching
- Loan Details
- Lender Decision

Payment Service

- Fund Transfer
- Repayments
- Payment Gateway
- Transaction Status
- Refunds

Notification Service

- Email / SMS
- In-App Alerts
- Reminders
- Loan Updates

Admin Service

- System Config
- Risk & Fraud Monitoring
- Reports
- Audit Logs

4. AI / CREDIT RISK ENGINE

Input Data

- Income
- Employment
- Existing Liabilities
- Credit History
- Repayment History
- Loan Amount
- Loan Tenure
- Other Features

Data Preprocessing

- Data Cleaning
- Missing Values
- Outlier Handling
- Encoding
- Normalization
- Feature Engineering

ML Models

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Model Training & Evaluation

Risk Score Output

- Default Probability
- Risk Score (0-100)
- Risk Category (Low / Medium / High)

Explainable AI

- Feature Importance
- SHAP Values
- Reason Codes
- Risk Insights for Lenders

5. DATA & STORAGE LAYER

Operational Database (PostgreSQL)

- Users
- Borrower Profiles
- Loans
- Loan Funding
- Repayments
- Risk Assessments
- Transactions
- Audit Logs

File Storage

- KYC Documents
- Loan Documents
- Agreement Files
- Other Files

Cache (Redis)

- Session Store
- Frequently Accessed Data
- OTP / Token Store
- Rate Limiting

6. BLOCKCHAIN LAYER (TRANSPARENCY & AUDIT TRAIL)

Smart Contract (Solidity)

- createLoan() – Create loan record
- fundLoan() – Record funding event
- recordRepayment() – Record repayment
- updateLoanStatus() – Update loan status
- closeLoan() – Close loan

1. Loan Created

Loan details hashed & stored on-chain

2. Loan Funded

Funding event recorded on-chain

3. Repayment Event

Each repayment recorded on-chain

4. Loan Completed

Loan closure event stored on-chain

Immutable Audit Trail → Transaction Hash → Transparent & Tamper Proof Records

EXTERNAL SERVICES

KYC / ID Verification Provider

- Identity Verification
- Document Verification
- Address Verification

Payment Gateway / Bank Integration

- Loan Funding
- Repayments
- Payouts / Settlements
- Webhook / Status

Notification Provider

- Email Service
- SMS Service
- Push Notifications

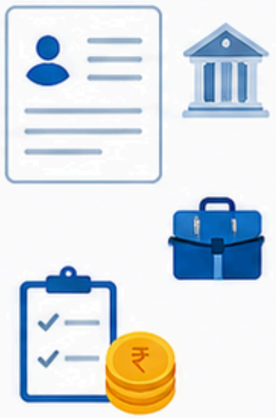
Blockchain Network (Polygon Testnet)

- Smart Contract Deployment
- Transaction Broadcast
- Event Listening

Methodology/Algorithm

BorroWealth – AI Powered P2P Lending Platform

1 Data Collection



Collect borrower financial, employment, loan and repayment data.

2 Data Preprocessing



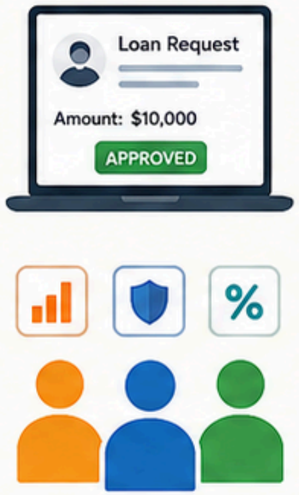
Clean, transform and prepare the data for AI analysis.

3 AI Credit Scoring



ML model analyzes borrower characteristics and generates a risk score and risk category.

4 P2P Lending Marketplace



Approved loan requests are listed for lenders, who evaluate risk insights and choose opportunities.

5 Blockchain Integration



Smart contracts record key loan events such as funding and repayments, providing transparency and auditability.

6 Platform Development



Integrate the AI model, marketplace, database, authentication and blockchain into the BorroWealth application.

7 Testing & Evaluation



Evaluate AI performance, application functionality, security and blockchain transactions.

Dataset Details

Lending Club + Home Credit

The Lending Club data includes loan and borrower information such as income, employment, debt-to-income ratio, loan characteristics and credit history, while Home Credit combines the main application table with bureau records, previous applications, credit-card history and installment-payment history.

Lending Club (Primary Development)

- Large-scale loan records
- Clear default prediction task
- Strong fit for XGBoost
- Supports class-imbalance research
- Main source for risk-tier prediction

Home Credit (Robustness & Validation)

- Multi-source borrower data
- Rich credit history
- Advanced behavioural features
- Complex feature engineering
- Tests model methodology across another lending environment

The combination of both datasets strengthens the project by demonstrating that the proposed AI-driven credit risk methodology is not limited to a single lending environment.

TWO DATASETS • TWO LENDING ENVIRONMENTS • ONE ROBUST CREDIT-RISK FRAMEWORK

Implementation plan for both semesters

References

https://docs.google.com/document/d/1HQaPVPiKeEoU2Zvcg3BypSwnpXHHO6htiBkvYwHB5Iw/edit?tab=t.0



THANK YOU